Quadratics: Expanding & Factorising

Practice test 7 — answer sheet · 40 marks

Section A — Expanding double brackets

A1. $x^2 - 2x + 7x - 14$ (1) = $\mathbf{x^2 + 5x - 14}$ (1)

A2. $x^2 - 12x - x + 12$ (1) = $\mathbf{x^2 - 13x + 12}$ (1)

A3. $6x^2 - 2x - 21x + 7$ (1) = $\mathbf{6x^2 - 23x + 7}$ (1)

A4. $(2x + 3)(2x + 3) = 4x^2 + 6x + 6x + 9$ (1) = $\mathbf{(4x^2 + 12x + 9)\ cm^2}$ (1)

Section B — Expanding three brackets

B1. $(x + 1)(x + 4) = x^2 + 5x + 4$ (1) $(x^2 + 5x + 4)(x - 5) = x^3 - 5x^2 + 5x^2 - 25x + 4x - 20$ (1) = $\mathbf{x^3 - 21x - 20}$ (1)

B2. $(2x - 1)^2 = 4x^2 - 4x + 1$ (1) $(4x^2 - 4x + 1)(x + 3) = 4x^3 + 12x^2 - 4x^2 - 12x + x + 3$ (1) = $\mathbf{4x^3 + 8x^2 - 11x + 3}$ (1)

Section C — Factorising $x^2 + bx + c$ and the difference of two squares

C1. Multiply to 45, add to 14: 5 and 9 (1) **(x + 5)(x + 9)** (1)

C2. Multiply to –63, add to –2: –9 and 7 (1) **(x – 9)(x + 7)** (1)

C3. $7^2 - (2x)^2$ = **(7 + 2x)(7 – 2x)** (1)

C4. –4 × a = –20 so **a = 5** (1) b = –4 + 5, so **b = 1** (1)

Section D — Factorising $ax^2 + bx + c$

D1. 3 × 8 = 24; numbers: 12 and 2, so $3x^2 + 12x + 2x + 8 = 3x(x + 4) + 2(x + 4)$ (1) **(3x + 2)(x + 4)** (1)

D2. 5 × 4 = 20; numbers: –10 and –2, so $5x^2 - 10x - 2x + 4 = 5x(x - 2) - 2(x - 2)$ (1) **(5x – 2)(x – 2)** (1)

D3. $2(6x^2 + 11x - 2)$ (1) = **2(6x – 1)(x + 2)** (1)

Section E — Using factorising

E1. $(2x - 1)(x - 3) = 0$ (1) $2x - 1 = 0$ or $x - 3 = 0$ (1) **x = 0.5 or x = 3** (1)

E2. Numerator = $(x - 5)(x + 2)$ (1) denominator = $(x + 2)(x - 2)$ (1) cancel (x + 2): **(x – 5) / (x – 2)** (1)

Section F — Problem solving & proof

F1. (a) $(x + 3)^2 = x^2 + 6x + 9$ (1) so $x^2 + 6x - 4$ = $\mathbf{(x + 3)^2 - 13}$ (1)
(b) a square can't be negative, so the **minimum value is** –**13** (when x = –3) (1)

F2. Let the numbers be 2n and 2n + 2 (1) $2n(2n + 2) + 1 = 4n^2 + 4n + 1$ (1) = $(2n + 1)^2$, which is **a square number** (1)

F3. (76 + 24)(76 – 24) = 100 × 52 = **5200** (1)

Marking note: the marks in brackets show where each mark is earned. Award method marks for a correct method even if there is a later slip. Accept answers in any equivalent form, e.g. terms in a different order, or brackets written the other way round: $(x – 7)(x + 4)$ is the same as $(x + 4)(x – 7)$. “Factorise fully” questions lose the last mark if a common factor is left inside a bracket.